Internet Crimes and Cyber Security

Week: 04 **Session: 16**

Topic :Introduction to Cyberspace & Cyber Security

B Md Irfan
Technical Coordinator (Cyber Security)
NALSAR University of Law, Hyderabad

Introduction to Cyberspace & Cyber Security

1. Introduction to Cyberspace

Cyberspace is a global, interconnected network of digital environments made up of the internet, computer systems, software, and data. It is the virtual space where digital communication and online activities take place.

2. Fundamentals of Cyber Security

Cyber security is the process of using best practices to protect computers, servers, systems, networks, and programs from digital attacks such as viruses, worms, ransomware, and other threats aimed at accessing, changing, or destroying sensitive data, extorting money from users, or interrupting normal business processes.

The main idea behind cyber security is:

1. Identify and fix security vulnerabilities.

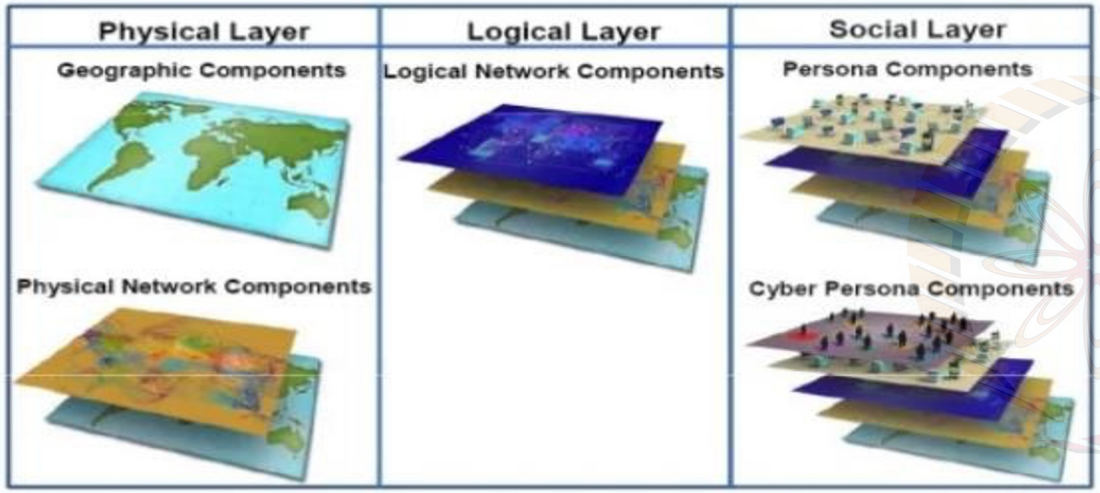
1. Prevent unauthorized access and data breaches.

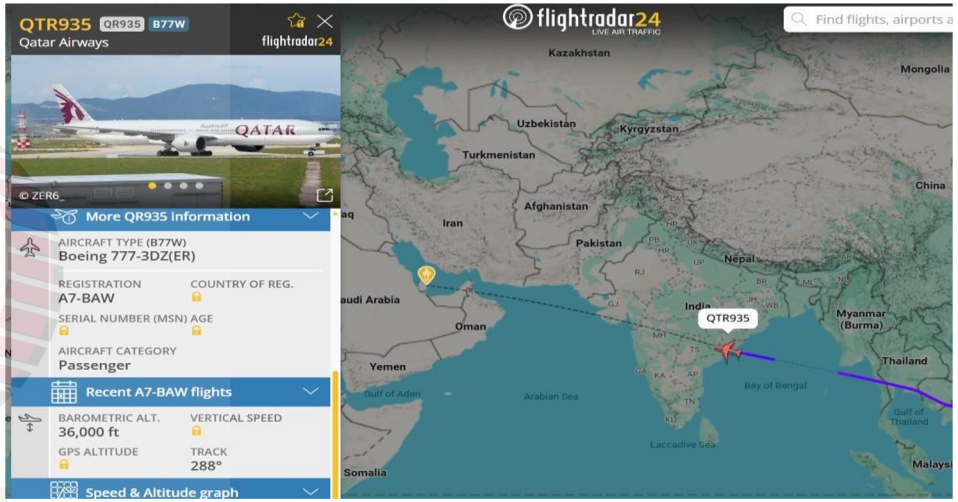
1. Protect systems from malicious attacks and disruptions.

1. Ensure confidentiality, integrity, and availability of information

Introduction to Cyberspace & Cyber Security

3. Three layers of Cyberspace





Importance of Cyberspace

- Facilitates global communication and data exchange
- Drives modern commerce and economy (e-commerce, online banking)
- Supports education, research, and government services
- Plays a role in social interaction (social media, forums)

4. Cyber Security Principles

- **Confidentiality:** Ensuring information is accessible only to authorized parties.
- **Integrity:** Ensuring data is accurate and unaltered.
- **Availability:** Ensuring data and services are accessible when needed.

The CIA Triad		
What Is the CIA?		
Confidentiality	Integrity	Availability
The information is safe from accidental or intentional disclosure.	The information is safe from accidental or intentional modification or alteration.	The information is available to authorized users when needed.
Example		
I send you a message, and no one else knows what that message is.	I send you a message, and you receive exactly what I sent you (without any modification)	I send you a message, and you are able to receive it.
What's The Purpose of the CIA?		
Data is not disclosed	Data is not tampered	Data is available
How Can You Achieve the CIA?		
e.g., Encryption	e.g., Hashing, Digital signatures	e.g., Backups, redundant systems
Opposite of CIA		
Disclosure	Alteration	Destruction

5. What is Cyber security Triad

The cybersecurity triad is a foundational model in information security, consisting of **confidentiality, integrity, and availability (CIA)**, which provides a guide for

1. **creating policies and security controls. Confidentiality ensures information is not accessed by unauthorized individuals, integrity protects data from unauthorized alteration, and availability guarantees that systems and data are accessible to authorized users when needed.** This model is used to assess and
2. **improve an organization's security posture and is referenced in many risk management frameworks.**

Introduction to Cyberspace & Cyber Security

6. Why Cyber Security is Important?

Protection against cyber-attacks (e.g., data breaches, malware)
Safeguarding privacy of individuals and organizations
Ensuring national security and critical infrastructure
Building trust in digital platforms

6. Key Components of Cyber Security

Network Security: Securing network infrastructure.
Information Security: Ensuring the integrity and privacy of data.
Application Security: Protecting software applications from cyber threats.
Endpoint Security: Protecting end-user devices (e.g., computers, smartphones).

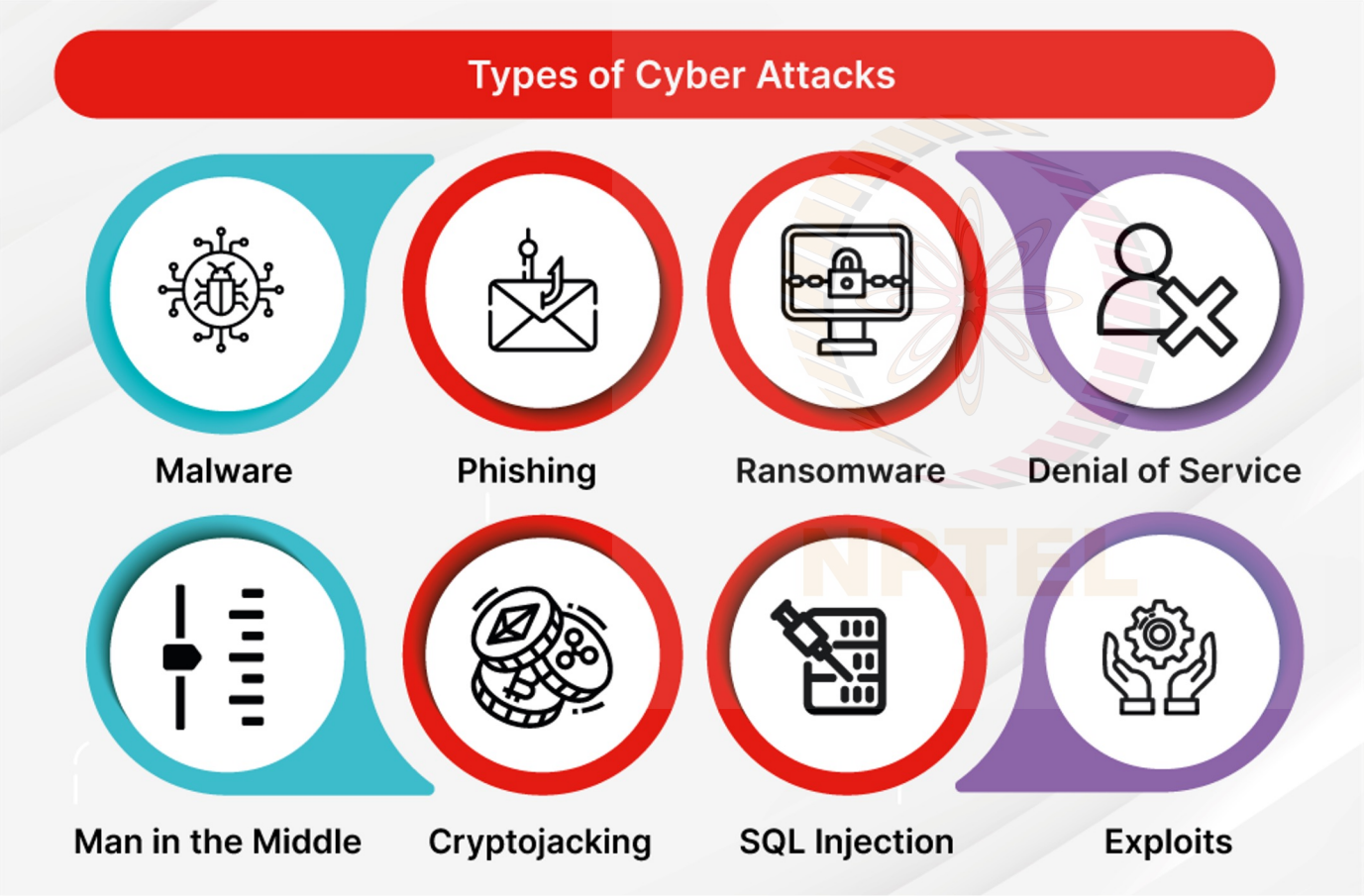
8. What are a Cyber Attack

A **cyber attack** is any deliberate attempt by an individual or group to breach the information system of another entity. These attacks aim to steal, expose, alter, disable, or destroy data, systems, or infrastructure.

Types of Cyber Threats

- **Malware**: Viruses, worms, ransomware, Trojans.
- **Phishing**: Deceptive emails designed to steal data.
- **Denial of Service (DoS)**: Attacks that disrupt services.
- **Man-in-the-Middle (MitM)**: Intercepting communication between parties.
- **Advanced Persistent Threats (APT)**: Prolonged, targeted attacks.

Introduction to Cyberspace & Cyber Security



G. Cyber Attack Lifecycle

- **Reconnaissance:** Information gathering about the target.
- **Weaponization:** Creating malware or exploits.
- **Delivery:** Sending the malicious payload.
- **Exploitation:** Exploiting vulnerabilities in the system.
- **Installation:** Installing malware on the victim's system.
- **Command s Control (CsC):** Establishing communication with compromised system.
- **Actions on Objectives:** Achieving the attacker's goals.

10. Cyber Security Strategies

- **Prevention**: Implementing firewalls, antivirus software, and encryption.
- **Detection**: Using intrusion detection systems (IDS) and anomaly detection.
- **Response**: Incident response plans and recovery procedures.
- **Recovery**: Backup systems and business continuity plans.

Thank you